

Feature Engineering using Spark, Hyperparameter Tuning using Ray Tune, and Experiment Tracking using MLflow

0. Version Used

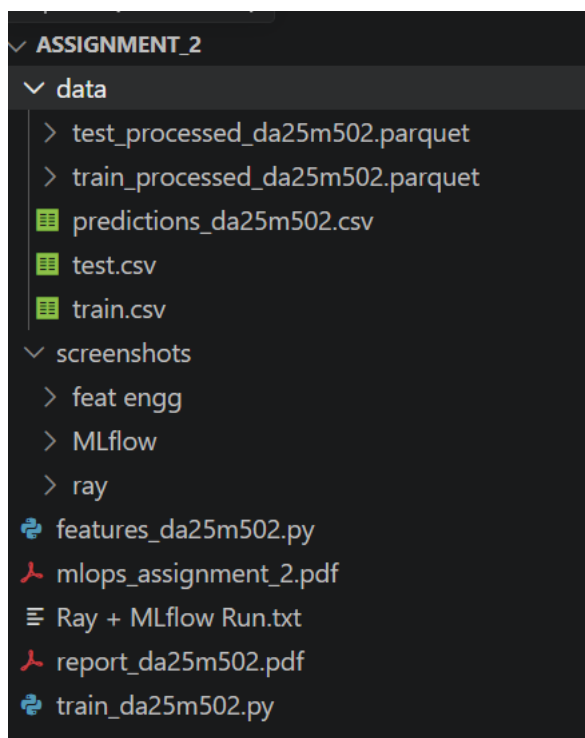
Spark: 3.5.6

Ray : 2.56.1

MLflow : 3.14.0(client)

MLflow url : <http://localhost:5000>

0.1 Folder Structure



1. Spark preprocessing pipeline

Script name: features_da25m502.py

```
(mlops_learning_project) sjagkoo7@DESKTOP-91C1HV8: ~/projects/MLOPS_Lab/Assignment_2$ python features_da25m502.py
26/07/19 09:28:55 WARN Utils: Your hostname, DESKTOP-91C1HV8 resolves to a loopback address: 127.0.1.1; using 10.255.255.254
instead (on interface lo)
26/07/19 09:28:55 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/07/19 09:28:56 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classe
s where applicable
Spark version      : 3.5.6
Master             : local[*]
Parallel workers   : 4
26/07/19 09:29:53 WARN SparkStringUtils: Truncated the string representation of a plan since it was too large. This behavior
can be adjusted by setting 'spark.sql.debug.maxToStringFields'.
Pipeline completed successfully! Transformed files saved.
```

Configuration

```
def main():  
    # 1. Initialize Spark Session and read datasets  
    spark = (  
        SparkSession.builder  
            .appName("features_da25m502_pipeline")  
            .config("spark.driver.memory", "2g")  
            .config("spark.sql.shuffle.partitions", "4")  
            .master("local[*]")  
            .getOrCreate()  
    )
```

Pipeline Stages

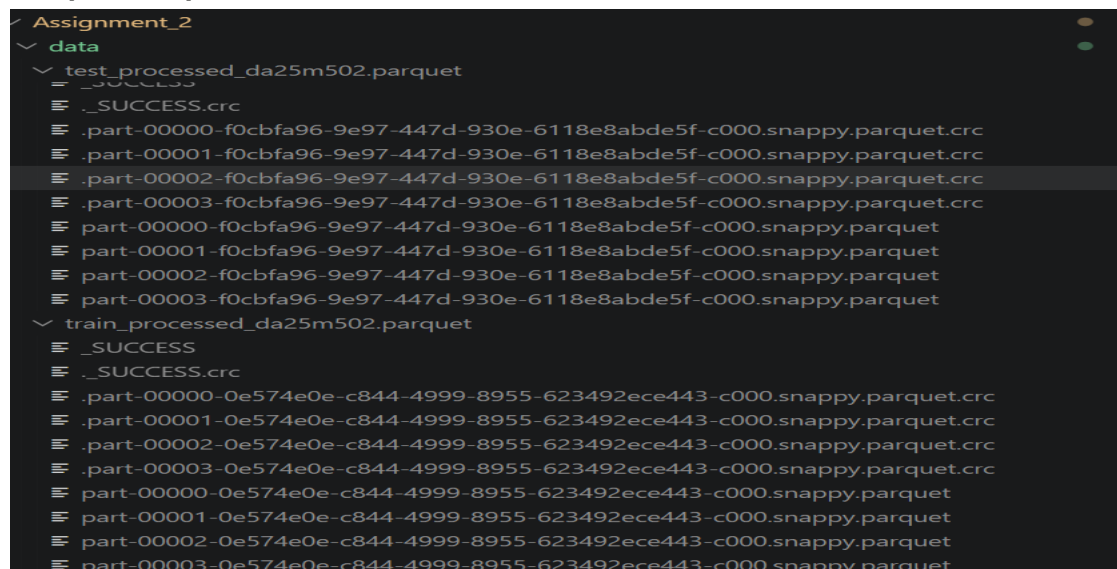
[Input Data] → [Imputation (Median/Mode)] → [StringIndexer] → [VectorAssembler] → [StandardScaler] → [Parquet Output]

StringIndexer : Converts string fields into ordered index integers based on frequency.

VectorAssembler : Combines numerical metrics and indexed labels into a unified multi-dimensional array vector.

StandardScaler : Normalizes values to achieve zero-mean and unit variance, preparing the final feature matrix.

Parquet Output:



```
Assignment_2  
└─ data  
  └─ test_processed_da25m502.parquet  
    ├── _SUCCESS  
    ├── _SUCCESS.crc  
    ├── .part-00000-f0cbfa96-9e97-447d-930e-6118e8abde5f-c000.snappy.parquet.crc  
    ├── .part-00001-f0cbfa96-9e97-447d-930e-6118e8abde5f-c000.snappy.parquet.crc  
    ├── .part-00002-f0cbfa96-9e97-447d-930e-6118e8abde5f-c000.snappy.parquet.crc  
    ├── .part-00003-f0cbfa96-9e97-447d-930e-6118e8abde5f-c000.snappy.parquet.crc  
    ├── part-00000-f0cbfa96-9e97-447d-930e-6118e8abde5f-c000.snappy.parquet  
    ├── part-00001-f0cbfa96-9e97-447d-930e-6118e8abde5f-c000.snappy.parquet  
    ├── part-00002-f0cbfa96-9e97-447d-930e-6118e8abde5f-c000.snappy.parquet  
    └── part-00003-f0cbfa96-9e97-447d-930e-6118e8abde5f-c000.snappy.parquet  
  └─ train_processed_da25m502.parquet  
    ├── _SUCCESS  
    ├── _SUCCESS.crc  
    ├── .part-00000-0e574e0e-c844-4999-8955-623492ece443-c000.snappy.parquet.crc  
    ├── .part-00001-0e574e0e-c844-4999-8955-623492ece443-c000.snappy.parquet.crc  
    ├── .part-00002-0e574e0e-c844-4999-8955-623492ece443-c000.snappy.parquet.crc  
    ├── .part-00003-0e574e0e-c844-4999-8955-623492ece443-c000.snappy.parquet.crc  
    ├── part-00000-0e574e0e-c844-4999-8955-623492ece443-c000.snappy.parquet  
    ├── part-00001-0e574e0e-c844-4999-8955-623492ece443-c000.snappy.parquet  
    ├── part-00002-0e574e0e-c844-4999-8955-623492ece443-c000.snappy.parquet  
    └── part-00003-0e574e0e-c844-4999-8955-623492ece443-c000.snappy.parquet
```

2. Hyperparameter search space

Here I have used the logistic regression, below is hyperparameter search space.

```

search_space = {
    "C": tune.loguniform(0.01, 10.0),
    "penalty": tune.choice(["l2"]),
    "solver": tune.choice(["lbfgs"]),
    "max_iter": tune.choice([20, 30, 50])
}

print("Beginning 10 Hyperparameter Tuning Trials via Ray Tune...")

```

Trial status: 10 PENDING
 Current time: 2026-07-21 18:44:10. Total running time: 0s
 Logical resource usage: 1.0/1 CPUs, 0/0 GPUs

Trial name	status	C	penalty	solver	max_iter
train_evaluate_lr_29199_00000	PENDING	3.58379	l2	lbfgs	50
train_evaluate_lr_29199_00001	PENDING	0.0258368	l2	lbfgs	50
train_evaluate_lr_29199_00002	PENDING	1.03525	l2	lbfgs	20
train_evaluate_lr_29199_00003	PENDING	0.0837697	l2	lbfgs	50
train_evaluate_lr_29199_00004	PENDING	0.458772	l2	lbfgs	50
train_evaluate_lr_29199_00005	PENDING	4.26228	l2	lbfgs	30
train_evaluate_lr_29199_00006	PENDING	7.35538	l2	lbfgs	50
train_evaluate_lr_29199_00007	PENDING	0.0769137	l2	lbfgs	20
train_evaluate_lr_29199_00008	PENDING	3.41451	l2	lbfgs	20
train_evaluate_lr_29199_00009	PENDING	0.0107596	l2	lbfgs	30

3.Best hyperparameters

Trial name	status	C	penalty	solver	max_iter	iter	total time (s)
train_evaluate_lr_29199_00000 0.512034	TERMINATED	3.58379	l2	lbfgs	50	1	12.5966
train_evaluate_lr_29199_00001 0.511669	TERMINATED	0.0258368	l2	lbfgs	50	1	10.1041
train_evaluate_lr_29199_00002 0.511983	TERMINATED	1.03525	l2	lbfgs	20	1	10.3506
train_evaluate_lr_29199_00003 0.51185	TERMINATED	0.0837697	l2	lbfgs	50	1	10.5099
train_evaluate_lr_29199_00004 0.511983	TERMINATED	0.458772	l2	lbfgs	50	1	10.6448
train_evaluate_lr_29199_00005 0.512034	TERMINATED	4.26228	l2	lbfgs	30	1	11.2885
train_evaluate_lr_29199_00006 0.512034	TERMINATED	7.35538	l2	lbfgs	50	1	10.8975
train_evaluate_lr_29199_00007 0.511867	TERMINATED	0.0769137	l2	lbfgs	20	1	10.9611
train_evaluate_lr_29199_00008 0.512034	TERMINATED	3.41451	l2	lbfgs	20	1	10.785
train_evaluate_lr_29199_00009 0.511338	TERMINATED	0.0107596	l2	lbfgs	30	1	10.7264

Overview	Traces	Artifacts
Status	Ready	
Model ID	m-b5fdecc63b2140faabd5d92434f59c33	
Source run	unequaled-bee-51	
Source run ID	a10d324b8161435384c1c47e275def1d	
Logged from	default_worker.py	
Datasets used	-	
Model versions	Assignment2Classifier_da25m502 v1 +1	

Metrics (8)			
Metric	Dataset	Source run	Value
recall	-	unequaled-bee-51	0.42119897675453233
roc_auc	-	unequaled-bee-51	0.926890923606018
training_time	-	unequaled-bee-51	0.5616617202738789
precision	-	unequaled-bee-51	0.6528184795724875
accuracy	-	unequaled-bee-51	0.90376
F1_score	-	unequaled-bee-51	0.5120335316387237

Parameters (4)	
Parameter	Value
penalty	l2
solver	lbfgs
max_iter	50
C	3.5837867720396055

4.Final model performance

```
Tuning Complete!
Highest Validation F1-Score: 0.51203

Registering the top model from Run ID (a10d324b8161435384c1c47e275def1d) in Model Registry...
Successfully registered model 'Assignment2Classifier_da25m502'.
2026/07/21 18:46:40 WARNING mlflow.tracking.model_registry.fluent: Run with id a10d324b8161435384c1c47e275def1d has no arti
facts at artifact path 'model', registering model based on models:/m-b5fdecc63b2140faabd5d92434f59c33 instead
2026/07/21 18:46:40 INFO mlflow.store.model_registry.abstract_store: Waiting up to 300 seconds for model version to finish c
reation. Model name: Assignment2Classifier_da25m502, version 1
Created version '1' of model 'Assignment2Classifier_da25m502'.
Model successfully registered under name: 'Assignment2Classifier_da25m502' as Version 1.

Loading 'Assignment2Classifier da25m502' version: 1 for verification...
Downloading artifacts: 100% | 5/5 [00:00<00:00, 92.81it/s]
Loading processed test dataset...
Executing batch inference predictions on transformed test features...
Success! Inference dataset generated and saved to: /home/sjagkoo7/projects/MLOPS_Lab/Assignment_2/data/predictions_da25m502.
.csv
```

Overview	Traces	Artifacts
Status	Ready	
Model ID	m-b5fdecc63b2140faabd5d92434f59c33	
Source run	unequaled-bee-51	
Source run ID	a10d324b8161435384c1c47e275def1d	
Logged from	default_worker.py	
Datasets used	-	
Model versions	Assignment2Classifier_da25m502 v1 +1	

Metrics (8)			
Metric	Dataset	Source run	Value
recall	-	unequaled-bee-51	0.42119897675453233
roc_auc	-	unequaled-bee-51	0.926890923606018
training_time	-	unequaled-bee-51	0.5616617202738789
precision	-	unequaled-bee-51	0.6528184795724875
accuracy	-	unequaled-bee-51	0.90376
F1_score	-	unequaled-bee-51	0.5120335316387237

Parameters (4)	
Parameter	Value
penalty	l2
solver	lbfgs
max_iter	50
C	3.5837867720396055

5.Generated predictions .csv file


predictions_da25m502.csv

	prediction_probability	prediction_class
1	0.014997849388748156	0
2	0.2526826736609584	0
3	0.004588971696313413	0
4	0.0025276553248124193	0
5	0.03238869221239061	0
6	0.05277958293257084	0
7	0.8102647489128507	1
8	0.3032556392569416	0
9	0.05099241367417999	0
10	0.3274169587823909	0
11	0.006209326157940758	0
12	0.5206303597862457	1
13	0.192642690443945	0
14	0.024843109387301877	0
15	0.1181457943808393	0
16	0.12588707442150873	0
17	0.006373193517964293	0
18	0.12572919504961258	0
19	0.1724377851267875	0
20	0.11100378614994665	0
21	0.26976227743373066	0
22	0.41496553283120086	0
23		

6. Inference process

Batch process

```
Tuning Complete!
Highest Validation F1-Score: 0.51203

Registering the top model from Run ID (a10d324b8161435384c1c47e275def1d) in Model Registry...
Successfully registered model 'Assignment2Classifier_da25m502'.
2026/07/21 18:46:40 WARNING mlflow.tracking.model_registry.fluent: Run with id a10d324b8161435384c1c47e275def1d has no artifacts at artifact path 'model', registering model based on models:/m-b5fdecc63b2140faabd5d92434f59c33 instead
2026/07/21 18:46:40 INFO mlflow.store.model_registry.abstract_store: Waiting up to 300 seconds for model version to finish creation. Model name: Assignment2Classifier_da25m502, version 1
Created version '1' of model 'Assignment2Classifier_da25m502'.
Model successfully registered under name: 'Assignment2Classifier_da25m502' as Version 1.

Loading 'Assignment2Classifier_da25m502' version: 1 for verification...
Downloading artifacts: 100% | 5/5 [00:00<00:00, 92.81it/s]
Loading processed test dataset...
Executing batch inference predictions on transformed test features...
Success! Inference dataset generated and saved to: /home/sjagkoo7/projects/MLOPS_Lab/Assignment_2/data/predictions_da25m502.csv
```

7. Screenshots of MLflow experiments and Model Registry



Ray + MLflow Run.txt

Experiment : assignment_2_da25m502 , total 10 runs

Run Name	Created	F1	Dataset	Duration	Source	Models
ray_tune_orchestrator	58 minutes ago	-	-	2.5min	train_da...	-
whimsical-donkey	55 minutes ago	-	-	10.6s	default...	model
beautiful-boar-2	56 minutes ago	-	-	10.7s	default...	model
lyrical-neut-986	56 minutes ago	-	-	10.9s	default...	model
delicious-wasp-	56 minutes ago	-	-	10.8s	default...	model
unique-crane-10	56 minutes ago	-	-	11.2s	default...	model
bright-stout-746	57 minutes ago	-	-	10.6s	default...	model
bouncy-owl-337	57 minutes ago	-	-	10.4s	default...	model
useful-cat-626	57 minutes ago	-	-	10.3s	default...	model
useful-lark-975	57 minutes ago	-	-	10.0s	default...	model
unregulated-bee-	58 minutes ago	-	-	12.5s	default...	Assignment2Classif...

Registered model : Assignment2Classifier_da25m502

localhost:5000/#/experiments/978723382330837962/models

mlflow 3.1.0 Experiments Models Prompts

Experiments

Search experiments

- Default
- assignment_2_da25m502

assignment_2_da25m502

Provide Feedback Add Description

Runs Models Experimental Evaluation Traces

metrics.rmse >= 0.8 Datasets Sort: Creation time Columns

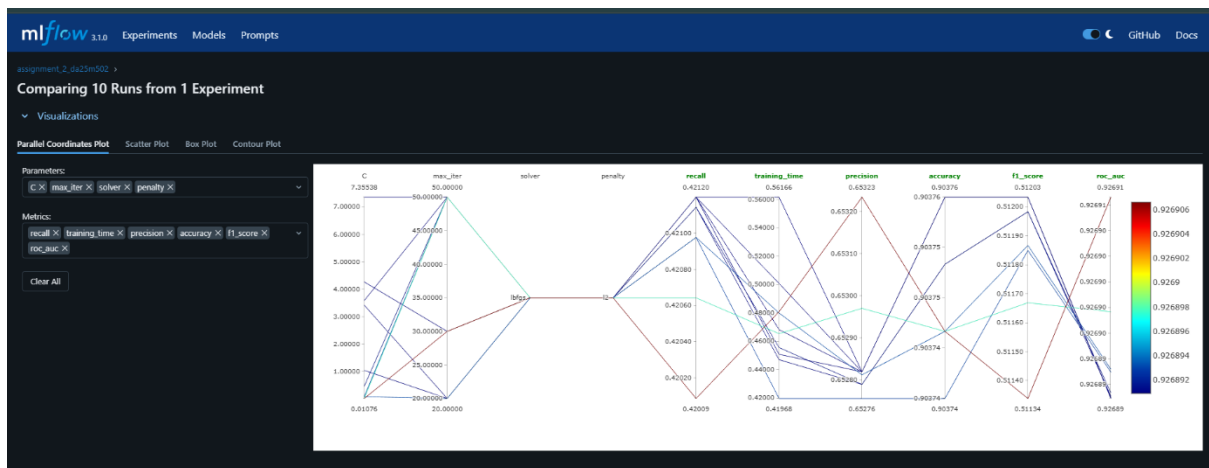
Model attributes			Model attributes			
Model name	Status	Created	Logged from	Source run	Registered models	Dataset
model	Ready	58 minutes ago	default_workor.py	whimsical-donkey-143	-	-
model	Ready	58 minutes ago	default_workor.py	beautiful-boar-270	-	-
model	Ready	58 minutes ago	default_workor.py	lyrical-newt-986	-	-
model	Ready	58 minutes ago	default_workor.py	debonair-wasp-890	-	-
model	Ready	59 minutes ago	default_workor.py	unique-crane-10	-	-
model	Ready	59 minutes ago	default_workor.py	bright-toast-746	-	-
model	Ready	59 minutes ago	default_workor.py	bouncy-owl-337	-	-
model	Ready	59 minutes ago	default_workor.py	useful-cat-626	-	-
model	Ready	1 hour ago	default_workor.py	useful-lark-975	-	-
model	Ready	1 hour ago	default_workor.py	unequaled-bee-51	Assignment2Classifier_da25m5	-

Model attributes			No dataset			
Model name	Status	Created	recall	roc_auc	training_time	precision
model	Ready	58 minutes ago	0.4200867534200868	0.9269066257212655	0.4808666706085205	0.6532341750259426
model	Ready	58 minutes ago	0.42119897675453233	0.9268909282396501	0.4508371353149414	0.6528184795724875
model	Ready	58 minutes ago	0.4209765320876432	0.9268931675476404	0.4799528121948242	0.6528113142462918
model	Ready	58 minutes ago	0.42119897675453233	0.926890884009526	0.46825075149536133	0.6528184795724875
model	Ready	59 minutes ago	0.42119897675453233	0.9268909143387539	0.5008342266082764	0.6528184795724875
model	Ready	59 minutes ago	0.42114336558781	0.92689132378333	0.4555950164794922	0.6527885527109732
model	Ready	59 minutes ago	0.4209765320876432	0.9268929447120636	0.4196796417236328	0.6527550228507373
model	Ready	59 minutes ago	0.42114336558781	0.9268910588238256	0.4470362663269043	0.6527885527109732
model	Ready	1 hour ago	0.4206428650873095	0.9268976680880642	0.46540212631225586	0.6529696132596685
model	Ready	1 hour ago	0.42119897675453233	0.926890923606018	0.5616617202758789	0.6528184795724875

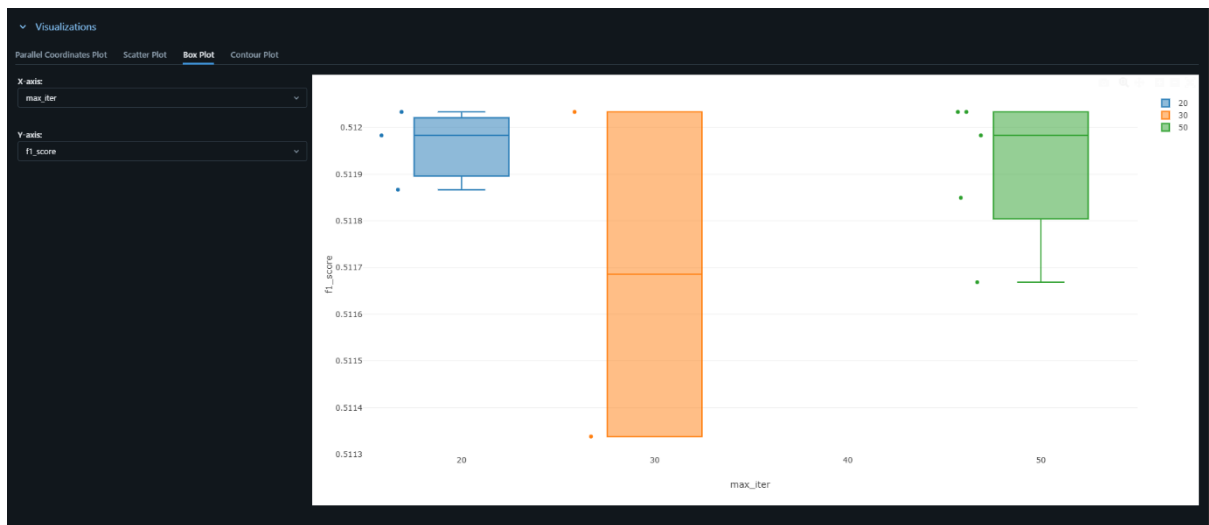
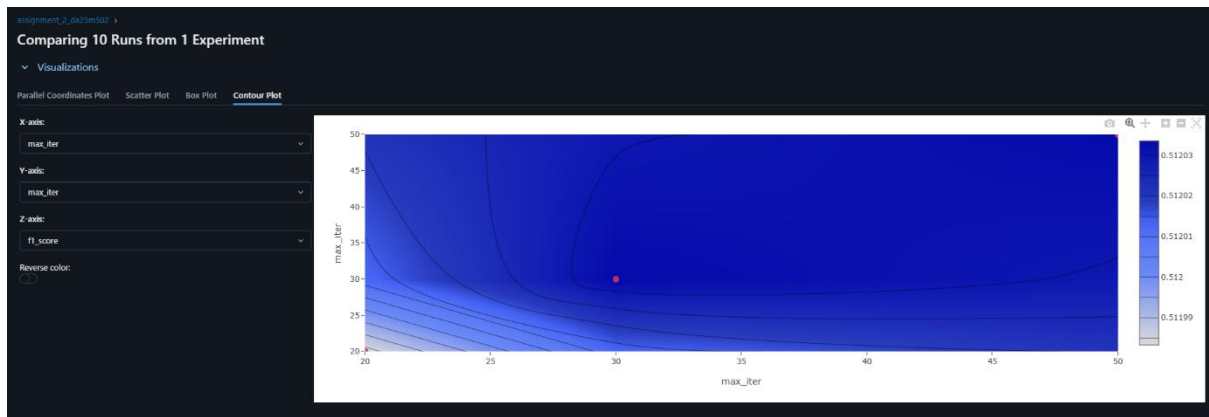
Model attributes			Parameters			
Model name	Status	Created	accuracy	f1_score	penalty	solver
model	Ready	58 minutes ago	0.9037466666666667	0.5113382522168821	l2	lbfgs
model	Ready	58 minutes ago	0.90376	0.5120335316387237	l2	lbfgs
model	Ready	58 minutes ago	0.9037466666666667	0.511866928122253	l2	lbfgs
model	Ready	58 minutes ago	0.90376	0.5120335316387237	l2	lbfgs
model	Ready	59 minutes ago	0.90376	0.5120335316387237	l2	lbfgs
model	Ready	59 minutes ago	0.9037533333333333	0.511983236138999	l2	lbfgs
model	Ready	59 minutes ago	0.90374	0.5118496230433753	l2	lbfgs
model	Ready	59 minutes ago	0.9037533333333333	0.511983236138999	l2	lbfgs
model	Ready	1 hour ago	0.9037466666666667	0.5116688087668267	l2	lbfgs
model	Ready	1 hour ago	0.90376	0.5120335316387237	l2	lbfgs

Model attributes				Parameters			
Model name	Status	Created		penalty	solver	max_iter	C
 model	 Ready	58 minutes ago	522168821	l2	lbfgs	30	0.010759576492531188
 model	 Ready	58 minutes ago	316387237	l2	lbfgs	20	3.4145092275157016
 model	 Ready	58 minutes ago	28122253	l2	lbfgs	20	0.07691369492724244
 model	 Ready	58 minutes ago	316387237	l2	lbfgs	50	7.355384215002141
 model	 Ready	59 minutes ago	316387237	l2	lbfgs	30	4.26227621958235
 model	 Ready	59 minutes ago	336138999	l2	lbfgs	50	0.45877183375217223
 model	 Ready	59 minutes ago	230433753	l2	lbfgs	50	0.08376974192965839
 model	 Ready	59 minutes ago	336138999	l2	lbfgs	20	1.0352498704446338
 model	 Ready	1 hour ago	287668267	l2	lbfgs	50	0.025836844682074908
 model	 Ready	1 hour ago	316387237	l2	lbfgs	50	3.5837867720396055

Model run comparison

[illegible]

Metrics									
☐ Show diff only									
accuracy	0.904	0.904	0.904	0.904	0.904	0.904	0.904	0.904	0.904
f1_score	0.511	0.512	0.512	0.512	0.512	0.512	0.512	0.512	0.512
precision	0.652	0.652	0.652	0.652	0.652	0.652	0.652	0.652	0.652
recall	0.42	0.421	0.421	0.421	0.421	0.421	0.421	0.421	0.421
roc_auc	0.927	0.927	0.927	0.927	0.927	0.927	0.927	0.927	0.927
training_time	0.481	0.451	0.48	0.468	0.501	0.456	0.42	0.447	0.465



Individual run

localhost:5000/#/experiments/978723382330837962/runs/45f77f07416140fc819d63874fa7239b

mlflow 3.1.0 Experiments Models Prompts

experiment_2_da75m502

lyrical-newt-986

Overview Model metrics System metrics Traces Artifacts

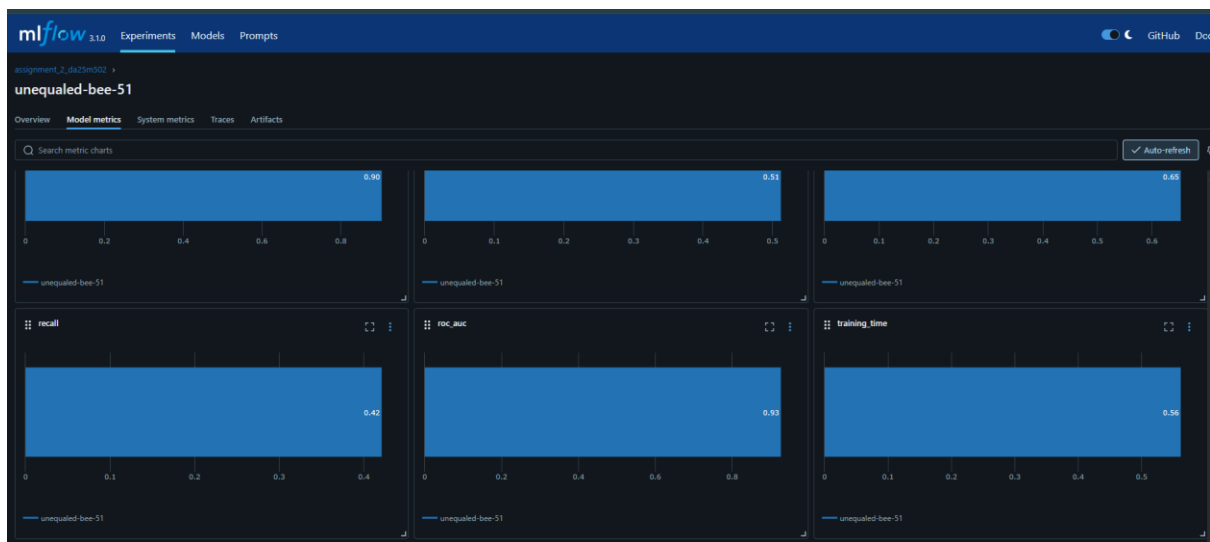
Experiment ID	978723382330837962
Status	Finished
Run ID	45f77f07416140fc819d63874fa7239b
Duration	10.9s
Parent run	ray_tune_orchestrator
Datasets used	—
Tags	Add tags
Source	default_worker.py
Registered models	—
Registered prompts	—

Metrics (6)

Metric	Value
recall	0.4209765320876432
roc_auc	0.0368931675476404
training_time	0.4799528121948242
precision	0.6028113142462918
accuracy	0.9037466666666667

Parameters (4)

Parameter	Value
penalty	l2
solver	lbfgs
max_iter	20
C	0.07691369492724244



mlflow 3.1.0 Experiments Models Prompts

assignment_2_da25ed02

unequaled-bee-51

Overview Model metrics System metrics Traces **Artifacts**

classification_report.txt 1268

Path: mlflow-artifacts/978723382330837962/a10d324b8161435384c1d7e275def1d/artifacts/classification_report.txt

	precision	recall	f1-score	support
0	0.92	0.97	0.95	132818
1	0.65	0.42	0.51	17982
accuracy				0.90 150800
macro avg	0.79	0.70	0.73	150800
weighted avg	0.89	0.90	0.89	150800